Gonzalo Díaz Amor

Data Scientist | Machine Learning Researcher | Computational Neuroscientist

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Professional Profile

Interdisciplinary Data Scientist and Neuroscientist, currently in the final stage of a Ph.D. in Computer Science (University of Buenos Aires). Specialized in **behavioral analysis, eye-tracking, and Machine Learning applied to computational psychiatry**. My experience ranges from predictive modeling and advanced processing of psychophysiological signals and trajectories to Open Source software development. Passionate about translating complex human behavior problems into data-driven solutions, communicating them effectively, and building valuable, scalable software products.

Relevant Experience

Graduate Researcher (Ph.D. Candidate in Computer Science)

Applied Artificial Intelligence Lab (LIAA, ICC, UBA-CONICET) | *2020 - 2026*

- **Project:** Development of computational tools for the early detection of cognitive deficits using predictions based on visual and manual trajectories.
- **Predictive Modeling:** Application of Machine Learning to classify and understand the algorithm behind complex task execution in patients. Feature engineering derived from high-precision measurements (hand & eye tracking).
- **Experimental Web Development:** Implementation of remote cognitive assessments via webcam eye-tracking, developing custom browser-based data capture solutions.
- *Funding: CONICET Doctoral Fellowship.*

Data Scientist

Development Bank of Latin America and the Caribbean (CAF) | *2021 - 2022*

- **Project:** Early warning system to prevent school dropout in the province of Mendoza, Argentina.
- **Tasks:** Data extraction and cleaning, creation and evaluation of Machine Learning models (risk classification/prediction), and development of dashboards for decision-making.

Open Source Maintainer & Core Contributor

DataPruebas / "pyxations" Library (Python) | *2024 - Present*

- **DataPruebas:** Co-creator of datapruebas.org, an online platform for scientific data collection, integrating open-source frameworks (jsPsych, PsychoPy) and enabling multi-modal data capture (eye-tracking, audio, mouse).
- **Pyxations:** Co-creator and core contributor of Pyxations, a Python library designed to process, parse, and analyze eye-tracking events and data.
- Software architecture design, implementation of best practices, and open-source project management.

Data Analyst (UBA Extension Volunteer)

Popular Data Workshop (Taller de Datos Populares) - Exactas UBA | *2022 - 2024*

- Application of Data Science tools to address social issues.
- Development of **Web Scraping** pipelines to extract, analyze, and consolidate a database of short-term rental prices, assessing their economic impact in the Argentine Patagonia.

Data Science & Programming Instructor

NeuroTransmitiendo, Instituto Humai, UBA, Instituto CPE & Argentine Society for Research in Neurosciences (SAN) | *2021 - Present*

- **Machine Learning Instructor:** Teacher in the Cognitive Neurosciences Diploma (NeuroTransmitiendo). Teaching Assistant for "Thesis Workshop I" (Master's in Data Mining and Knowledge Discovery, UBA).
- **Programming Instructor:** Teacher and material developer for Python courses (**Instituto Humai**). Instructor of *Data Wrangling* and *Data Visualization with Python* (**Instituto CPE**, remote, Uruguay), and the "Data Analysis and Visualization with Python" course (**SAN** 2022).

Educational Content Creator & Tech Writer

freeCodeCamp & Independent Publications | *2021 - Present*

- **freeCodeCamp:** First external contributor to the Spanish YouTube channel. Creator of the comprehensive Web Scraping with Python and BeautifulSoup course.
- **Outreach:** Active author of technical articles on Medium and Substack about interactive Python, NLP (e.g., semantic search engines with Streamlit), and programming tutorials.

Education

- **Ph.D. in Computer Science** | UBA (2020 - 2026)
  - Topic: *Computational psychiatry: individual differences and identification of executive function deficits.*
- **B.S. in Biological Sciences (Neurosciences track)** | UBA (2011 - 2019)
  - GPA: 7.96 / 10. Thesis conducted at the Applied Artificial Intelligence Lab.

Notable Additional Education (see LinkedIn for more):

- **Generative Artificial Intelligence** (UBA, 2023): Language Models, Word Embeddings, Deep Learning, Computer Vision.
- **Data Lab** (UBA, 2021): NLP, Web Scraping, APIs, Advanced Regression/Classification Models. *Distinction: Best predictive model in the course's internal Kaggle-style competition.*
- **Neuromatch Academy** (2020): Deep Learning, GLM, Dimensionality Reduction, Bayesian Statistics.

Mentorship and Co-direction of Theses (B.S. in Computer Science, UBA):

- **Francisco Figari:** *Applicability of remote web eye-tracking to neuropsychological diagnosis* (Defended 2022).
- **Agustín Penas:** *Automatic detection of eye-tracking events* (In progress).
- **Gianluca Capelo:** *Prediction of individual differences in executive functions using ML* (In progress).

Selected Scientific Production

(For the complete list of papers and posters refer to Google Scholar)

1. Verón, G. L., **Juantorena, G. E.**, Keller, G., Crivelli, L., & Kamienkowski, J. E. (2026). *Eye tracking as a diagnostic tool in Alzheimer's disease, mild cognitive impairment, and related dementias: a systematic review.* In **Alzheimer's & Dementia: Diagnosis, Assessment & Disease Monitoring**, 18(1), e70238.
2. **Juantorena, G. E.**, Capelo, G., Leon Vallejos, B. D., Ibáñez, A., Petroni, A., Berrios, W., Fernández, M. C., & Kamienkowski, J. E. (2026). *Machine Learning Based Digital Assessment of Mild Cognitive Impairment Using Hand Movements during the Trail Making Test.* In **medRxiv**.
3. Belloli, L., Keller, G., Carello, A., Gauder, L., **Juantorena, G. E.**, Corvalan, N., Allegri, R. F., Crivelli, L., & Fernández Slezak, D. (2025). *Multi-modal AI screening for MCI and Alzheimer's Disease: results from an Argentine Cohort.* In **Alzheimer's & Dementia**.
4. **Juantorena, G. E.**, Gauder, L., Laciana, P., Ferrer, L., & Kamienkowski, J. E. (2025). *DataPruebas: An Online Platform for Data Collection.* In **Argentine Symposium on Artificial Intelligence and Data Science (ASAID 2025)-JAIIO 54**.
5. **Juantorena, G. E.**, Capelo, G., Leon Vallejos, B. D., Berrios, W., Fernández, M. C., Ibáñez, A., Petroni, A., & Kamienkowski, J. E. (2024). *Enhancing Cognitive Assessment: Integrating Hand and Eye Tracking in the Digital Trail-Making Test for Mild Cognitive Impairment.* In **ETRA '24** (ACM Symposium on Eye Tracking Research and Applications).
6. **Juantorena, G. E.**, Figari, F., Petroni, A., & Kamienkowski, J. E. (2023). *Web-based eye-tracking for remote cognitive assessments: The anti-saccade task as a case study.* In **bioRxiv**.
7. Linari, I.*, **Juantorena, G. E.***, Ibáñez, A., Petroni, A., & Kamienkowski, J. E. (2022). *Unveiling Trail Making Test: visual and manual trajectories indexing multiple executive processes.* In **Scientific Reports (Nature)**, 12(1), 14265. (*Shared first authorship).

Technical Skills

- **Programming Languages:** Python, JavaScript, R, Matlab, SQL.
- **Data Science & Machine Learning:** Scikit-Learn, Pandas, NumPy, Exploratory Data Analysis (EDA), Statistical Modeling.
- **Deep Learning & NLP:** Pytorch, LLMs.
- **Data Engineering & Analysis:** Web Scraping (BeautifulSoup, automation), APIs, large-scale data preprocessing.
- **Development Tools & Frameworks:** Git/GitHub, Jupyter Notebooks, Google Colab, Streamlit, jsPsych.
- **Domain Specialties:** Eye-tracking, Signal Analysis (EEG, temporal trajectories), Online Experiment Design.
- **Languages:** English (Advanced), Spanish (Native).

Buenos Aires, Argentina | Available for relocation/remote work pending convenience.